

The Exact Fourier-Decay Theory of Rvachev's Up-Function

Reconciliation of four proposed solutions, correction of the lacunary-product LIL, and a complete resolution of the Cesàro normalization problem

Final synthesis for Vladimir Reshetnikov's ProveIt project

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Abstract

Let

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^n}\right), \quad \text{sinc } u = \frac{\sin u}{u},$$

be the Fourier transform of Rvachev's compactly supported up-function, with Fourier convention $\widehat{h}(\xi) = \int_{\mathbb{R}} h(t) e^{-2\pi i t \xi} dt$. The product is rapidly decreasing but has zeros at every nonzero integer and contains a lacunary dynamical factor. Consequently there is no single pointwise “decay exponent”: typical values, arithmetic means, quadratic means, fixed-ray extremals, dyadic-annulus maxima, and two-adically exceptional lobes have genuinely different lower-order behavior.

This article audits the four wave-1 analyses, the paper of Aistleitner–Hofer–Larcher, and the wave-2 comparative report, and then closes the remaining logical gap. The exact dyadic factorization reduces the problem to $P_n(z) = \prod_{j=0}^{n-1} |\sin(\pi 2^j z)|$. Its arithmetic mean is governed by the Perron root $\rho = 0.661322602060565\dots$ (numerical; rigorously $0.66130 < \rho < 0.66135$) of an explicit transfer operator. Thus the natural arithmetic gauge is

$$g_1(x) = \exp\left(-\frac{(\log x)^2}{2 \log 2}\right) x^{-\kappa_1}, \quad \kappa_1 = \frac{1}{2} + \log_2 \frac{\pi}{\rho} = 2.748070014871334\dots$$

After multiplication by $A_1 = 0.09126612413159\dots$, this gauge answers the bounded form of the original question and gives limit one along dyadic endpoints. At unrestricted times its means approach a nonconstant continuous mantissa profile generated by a singular transfer-operator eigenmeasure. This proves impossibility for every gauge whose shell distortion relative to g_1 is bounded, or whose shell shape stabilizes; it includes all Hardy-field gauges of the correct scale.

The principal new theorem is that this obstruction is not an unrestricted nonexistence theorem. By equalizing the mass on successively finer dyadic partitions inside each shell, respecting the one-sided derivative constraint imposed by monotonicity, and then applying the Hoischen–Carleman theorem on simultaneous entire approximation of a function and its derivative, we construct a positive real-analytic, eventually strictly decreasing gauge g for which

$$\lim_{t \rightarrow \infty} \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g(x)} dx = 1$$

for every fixed t_0 . It can be chosen with $\log g(x) = -(\log x)^2/(2 \log 2) - \kappa_1 \log x + o(\log x)$. Any such exact gauge must, however, develop unbounded and increasingly fine within-shell distortion; no simple stabilized formula can do the job.

We also give the complete exponent spectrum $\kappa_0 = 3.15149612947\dots$ (typical), $\kappa_1 = 2.74807001487\dots$ (arithmetic), $\kappa_2 = \log_2(2\pi) = 2.65149612947\dots$ (RMS), and $\kappa_\infty = 2.35901487911\dots$ (uniform coarse envelope), prove the sharp Lambert- W asymptotic for the largest peak in a dyadic annulus, and rederive the central limit theorem and law of the iterated logarithm. The exact variance is $\pi^2/4$, not $\pi^2/2$; hence the published LIL constant in the cited lacunary-product paper is too large by $\sqrt{2}$.

Final answer in one paragraph. There is no universal pointwise equivalent for $|\Phi(x)|$. The simple analytic gauge $A_1 g_1$ gives the correct arithmetic scale, satisfies the two-sided bounded requirement, and has a nonconstant dyadic-mantissa Cesàro profile. Every bounded-distortion, stabilizing, or Hardy-field modification of that gauge fails to make the unrestricted Cesàro limit equal to one. However, the literal existence question has an affirmative answer: an adaptive non-elementary positive analytic decreasing gauge can be constructed so that the unrestricted Cesàro limit is exactly one. Its increasingly fine shell microstructure is not optional; it is precisely what escapes the singular measure obstruction.

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1 Problem, conventions, and the hierarchy of answers

The original question asks for a real-valued gauge g , eventually analytic, positive, and monotone decreasing, such that either

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g(x)} dx \longrightarrow 1 \quad (t \rightarrow \infty), \quad (1)$$

or at least

$$0 < A < \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g(x)} dx < B < \infty \quad (t \geq t_1). \quad (2)$$

The phrase “decay rate” is ambiguous before one specifies what is being averaged. Four elementary facts already force this distinction.

1. $\Phi(m) = 0$ for every nonzero integer m , with nonuniform multiplicity. Therefore no positive pointwise equivalent can hold on the full real axis.
2. On a fixed dyadic ray $x = 2^k y$, the remaining oscillation is a Birkhoff sum for the doubling map. Typical rays fluctuate on a Gaussian and iterated-logarithm scale.
3. The arithmetic and quadratic shell normalizations are governed by different transfer-operator eigenvalues.
4. The largest peak in $[2^k, 2^{k+1}]$ is a boundary-layer optimization: it begins near the zero at 2^k , then shadows the maximizing orbit $1/3 \leftrightarrow 2/3$. This produces a Lambert- W correction absent from fixed-ray asymptotics.

Write throughout

$$\lambda = \log 2, \quad E(x) = \exp\left(-\frac{(\log x)^2}{2\lambda}\right). \quad (3)$$

The following table gives the correct power correction attached to the common log-normal factor $E(x)$. The precise meaning of each row is developed below.

notion of normalized amplitude	product scale	power exponent κ
Lebesgue-typical dyadic ray / geometric mean	$\Lambda_0 = \frac{1}{2}$	$\kappa_0 = \frac{3}{2} + \log_2 \pi = 3.151496129472319 \dots$
arithmetic normalization	$\Lambda_1 = \rho$	$\kappa_1 = \frac{1}{2} + \log_2(\pi/\rho) = 2.748070014871334 \dots$
root-mean-square normalization	$\Lambda_2 = 2^{-1/2}$	$\kappa_2 = \log_2(2\pi) = 2.651496129472318 \dots$
uniform coarse envelope	$\Lambda_\infty = \sqrt{3}/2$	$\kappa_\infty = \frac{3}{2} + \log_2(\pi/\sqrt{3}) = 2.359014879111741 \dots$

The ordering

$$\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0$$

is the usual ordering of extremal, quadratic, arithmetic, and geometric behavior. It is not a disagreement among valid calculations; it is a spectrum of answers to inequivalent questions.

2 Exact dyadic factorization and zero geometry

2.1 The shell identity

For $x > 0$, write uniquely

$$x = 2^k y, \quad k \in \mathbb{Z}, \quad 1 \leq y < 2, \quad z = y - 1 \in [0, 1).$$

Define

$$P_n(z) = \prod_{j=0}^{n-1} |\sin(\pi 2^j z)|, \quad H(y) = \prod_{m=1}^{\infty} \left| \operatorname{sinc}\left(\frac{\pi y}{2^m}\right) \right|. \quad (4)$$

The product H is continuous on $[1, 2]$, positive on $[1, 2)$, and vanishes at 2.

Proposition 2.1 (exact dyadic-shell factorization). *For every integer $k \geq 0$, $1 \leq y \leq 2$, and $n = k + 1$,*

$$|\Phi(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-n} H(y) P_n(y - 1). \quad (5)$$

Proof. Separate the factors with indices $0 \leq m \leq k$ from the tail. In the finite part put $j = k - m$. Since 2^j is an integer,

$$|\sin(\pi 2^j y)| = |\sin(\pi 2^j (y - 1))|.$$

The finite denominators contribute

$$\prod_{j=0}^k \pi 2^j y = (\pi y)^{k+1} 2^{k(k+1)/2},$$

and the remaining factors are exactly $H(y)$. Multiplication gives (5). $\square$

The universal quadratic term is already visible in $2^{-k(k+1)/2}$. Everything delicate is contained in P_n , together with the competition between its orbit and the factor y^{-n} .

2.2 Canonical product, multiplicities, and one peak per lobe

The Euler product for sinc can be regrouped without convergence ambiguity because the resulting genus-zero product is absolutely convergent.

Proposition 2.2 (canonical zero product). *For every $z \in \mathbb{C}$,*

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2} \right)^{1+\nu_2(m)}. \quad (6)$$

Thus the zeros at $\pm m$ have multiplicity $1 + \nu_2(m)$.

Proof. Expand each factor as

$$\operatorname{sinc}\left(\frac{\pi z}{2^h}\right) = \prod_{r=1}^{\infty} \left(1 - \frac{z^2}{r^2 4^h} \right).$$

A positive integer m is representable as $r 2^h$ exactly for $0 \leq h \leq \nu_2(m)$, giving the stated multiplicity. Finally, $\sum_m (1 + \nu_2(m)) / m^2 < \infty$, so the regrouping is legitimate. $\square$

Theorem 2.3 (strict log-concavity between consecutive zeros). *For every integer $m \geq 1$, $|\Phi|$ has exactly one critical point in $(m, m + 1)$; it is the unique maximum of that lobe.*

Proof. Away from the zeros, termwise differentiation in (6) gives

$$\frac{d^2}{dx^2} \log |\Phi(x)| = -2 \sum_{r=1}^{\infty} (1 + \nu_2(r)) \frac{r^2 + x^2}{(r^2 - x^2)^2} < 0. \quad (7)$$

Hence the logarithmic derivative is strictly decreasing on every interzero interval. It tends to $+\infty$ at the left endpoint and $-\infty$ at the right endpoint, so it has one and only one zero. $\square$

This theorem is useful both conceptually and computationally: the lobe maxima can be located by one-dimensional root finding, and exceptional multiplicities can be analyzed locally.

3 Transfer operator and the arithmetic eigenvalue

3.1 The doubling map and its weighted Perron operator

Let

$$Tz = 2z \pmod{1}, \quad w(z) = |\sin \pi z|.$$

Then $P_n(z) = \prod_{j=0}^{n-1} w(T^j z)$. For $q > 0$, define

$$(\mathcal{L}_q h)(x) = \frac{1}{2} \left[\sin^q \left(\frac{\pi x}{2} \right) h \left(\frac{x}{2} \right) + \cos^q \left(\frac{\pi x}{2} \right) h \left(\frac{x+1}{2} \right) \right]. \quad (8)$$

A change of variables on the two inverse branches gives

$$\int_0^1 h(z) P_n(z)^q dz = \int_0^1 \mathcal{L}_q^n h(x) dx. \quad (9)$$

We write $\mathcal{L} = \mathcal{L}_1$.

Theorem 3.1 (Ruelle–Perron–Frobenius package for $q = 1$). *On the Banach space of Lipschitz functions on $[0, 1]$, the operator $\mathcal{L}$ has a simple positive leading eigenvalue ρ , a strictly positive Lipschitz eigenfunction h , and a positive eigenprobability ν , normalized by $\nu(h) = 1$, such that*

$$\rho^{-n} \mathcal{L}^n \varphi \longrightarrow h \nu(\varphi) \quad \text{uniformly on } [0, 1] \quad (10)$$

for every Lipschitz φ . Moreover

$$\rho = 0.661322602060565 \dots \quad \text{numerically}, \quad 0.66130 < \rho < 0.66135 \quad \text{rigorously}. \quad (11)$$

Proof architecture. The elementary Lasota–Yorke estimate has the form

$$\text{Lip}(\mathcal{L}\varphi) \leq \frac{\sqrt{2}}{4} \text{Lip}(\varphi) + C \|\varphi\|_\infty,$$

while $\|\mathcal{L}\varphi\|_\infty \leq 2^{-1/2} \|\varphi\|_\infty$. Thus the essential spectral radius on the Lipschitz space is at most $\sqrt{2}/4$. The exact second moment $\int_0^1 P_n^2 = 2^{-n}$, together with the Gelfond estimate proved in [section 9](#), gives

$$\int_0^1 P_n dz \geq \frac{\int_0^1 P_n^2 dz}{\|P_n\|_\infty} \geq \sqrt{\frac{3}{5}} \left(\frac{1}{\sqrt{3}} \right)^n.$$

Hence $\rho \geq 1/\sqrt{3} > \sqrt{2}/4$, so the leading spectrum is isolated. Positivity, topological exactness of the doubling map, and the fact that the weights vanish only at branch endpoints give irreducibility and simplicity. This yields (10) by the standard positive quasi-compact-operator argument. The rigorous enclosure in (11) is the one proved in the lacunary-product paper by iterating concave upper and lower ratios; the longer decimal is a stable numerical collocation value. $\square$

For later use set

$$H_0 = \int_0^1 h(x) dx. \quad (12)$$

Combining (9) and (10) gives

$$\rho^{-n} \int_0^1 \varphi(z) P_n(z) dz \longrightarrow H_0 \nu(\varphi) \quad (13)$$

for continuous φ ; the extension from Lipschitz to continuous functions follows by uniform approximation.

3.2 The scale dictionary

Suppose a product norm grows like Λ^n . The power correction needed to cancel it in the exact shell factorization is

$$\kappa(\Lambda) = \frac{\log \pi + \lambda/2 - \log \Lambda}{\lambda}. \quad (14)$$

Indeed, for

$$g_\kappa(x) = E(x)x^{-\kappa},$$

a direct calculation from (5) gives

$$\frac{|\Phi(2^k(1+z))|}{g_\kappa(2^k(1+z))} = \Lambda^{-n} W_\Lambda(z) P_n(z), \quad n = k+1, \quad (15)$$

where

$$W_\Lambda(z) = \frac{\Lambda}{\pi} H(1+z)(1+z)^{\kappa-1} \exp\left(\frac{\log^2(1+z)}{2\lambda}\right). \quad (16)$$

This identity, not a heuristic averaging argument, is the common source of all four exponents.

For the arithmetic problem, put

$$\kappa_1 = \kappa(\rho) = \frac{1}{2} + \log_2 \frac{\pi}{\rho}, \quad g_1(x) = E(x)x^{-\kappa_1}, \quad (17)$$

and abbreviate

$$W(z) = W_\rho(z) = \frac{\rho}{\pi} H(1+z)(1+z)^{\kappa_1-1} \exp\left(\frac{\log^2(1+z)}{2\lambda}\right). \quad (18)$$

Then

$$\frac{|\Phi(2^k(1+z))|}{g_1(2^k(1+z))} = \rho^{-n} W(z) P_n(z). \quad (19)$$

4 The natural arithmetic gauge and its singular phase profile

4.1 The shell constant and limiting measure

Define the finite measures

$$\alpha_k(dz) = \rho^{-(k+1)} W(z) P_{k+1}(z) dz, \quad 0 \leq z \leq 1. \quad (20)$$

By (13),

$$\alpha_k \implies \alpha, \quad \alpha(dz) = H_0 W(z) \nu(dz). \quad (21)$$

Set

$$A_1 = \alpha([0, 1]) = H_0 \nu(W) = 0.09126612413159 \dots \quad (\text{numerical}), \quad (22)$$

and normalize

$$\mu(dz) = \frac{W(z) \nu(dz)}{\nu(W)}, \quad F(z) = \mu([0, z]). \quad (23)$$

The following regularity fact is crucial.

Theorem 4.1 (the arithmetic eigenmeasure is continuous, singular, and full support). *The eigenmeasure ν , and hence μ , has no atoms, is singular with respect to Lebesgue measure, and gives positive mass to every nonempty open subinterval of $(0, 1)$.*

Proof. For an atom at z , the dual eigenrelation gives

$$\rho\nu\{z\} = \frac{1}{2} \sin(\pi z) \nu\{Tz\}. \quad (24)$$

By [theorem 9.1](#) below, $\rho \geq 1/\sqrt{3} > 1/2$. Iterating (24) therefore yields

$$\nu\{z\} \leq (2\rho)^{-n} \longrightarrow 0,$$

so ν is atomless.

Suppose an absolutely continuous eigencomponent exists, with density $r \in L^1$, $r \geq 0$. A change of variables in the dual relation gives

$$\sin(\pi z) r(Tz) = \rho r(z), \quad r(T^n z) = \frac{\rho^n}{P_n(z)} r(z). \quad (25)$$

Birkhoff's theorem and $\int_0^1 \log |\sin \pi z| dz = -\log 2$ imply

$$P_n(z) = \exp[-n(\log 2 + o(1))]$$

for almost every z . On the set $\{r > 0\}$, the right side of (25) therefore grows as $\exp[n(\log(2\rho) + o(1))]$. On the other hand, for every $0 < \varepsilon < \log(2\rho)$, Markov's inequality and invariance of Lebesgue measure give

$$\sum_{n \geq 1} \text{Leb}\{z : r(T^n z) > e^{\varepsilon n}\} = \sum_{n \geq 1} \text{Leb}\{r > e^{\varepsilon n}\} < \infty.$$

Borel–Cantelli contradicts the preceding exponential lower growth. The dual operator sends absolutely continuous measures to absolutely continuous measures and singular measures to singular measures, so uniqueness of the Lebesgue decomposition shows that the absolutely continuous part of ν is zero.

Finally, every nonempty open interval contains a dyadic cylinder on which some iterate of T maps onto $(0, 1)$, with positive product weight away from finitely many endpoints. Applying the dual eigenrelation and using atomlessness shows that its ν -mass is positive. Since $W > 0$ on $[0, 1]$, the same is true for μ . $\square$

4.2 Exact dyadic-mantissa Cesàro profile

Let

$$g_{\text{nat}}(x) = A_1 g_1(x). \quad (26)$$

The natural gauge is positive and real-analytic on $(0, \infty)$, and is strictly decreasing for all sufficiently large x .

Theorem 4.2 (natural-gauge profile). *Fix $t_0 > 0$. For every $1 \leq y \leq 2$,*

$$\lim_{k \rightarrow \infty} \frac{1}{2^k y - t_0} \int_{t_0}^{2^k y} \frac{|\Phi(x)|}{g_{\text{nat}}(x)} dx = \mathcal{P}(y), \quad (27)$$

where

$$\boxed{\mathcal{P}(y) = \frac{1 + F(y-1)}{y}}. \quad (28)$$

The convergence is uniform in $y \in [1, 2]$. The function $\mathcal{P}$ is continuous, satisfies $\mathcal{P}(1) = \mathcal{P}(2) = 1$, and is not identically one.

Proof. By (19), for $0 \leq z \leq 1$,

$$\int_{2^k}^{2^{k(1+z)}} \frac{|\Phi(x)|}{A_1 g_1(x)} dx = 2^k \frac{\alpha_k([0, z])}{A_1}. \quad (29)$$

Weak convergence to the atomless measure α implies uniform convergence of the corresponding distribution functions, hence

$$\frac{\alpha_k([0, z])}{A_1} \longrightarrow F(z)$$

uniformly. A full shell contributes $2^k(1 + o(1))$. Summing all preceding shells and using the geometric weights gives

$$\int_{t_0}^{2^{k(1+z)}} \frac{|\Phi|}{A_1 g_1} = 2^k(1 + F(z) + o(1)),$$

uniformly in z . Division by $2^k(1 + z) - t_0$ proves (28). If $\mathcal{P} \equiv 1$, then $F(z) = z$, contradicting singularity of μ . $\square$

Corollary 4.3 (answer to the bounded question). *The gauge g_{nat} satisfies (2). It also satisfies (1) along the dyadic sequence $t = 2^k$, but not for unrestricted t .*

Proof. The continuous function $\mathcal{P}$ has a positive minimum and finite maximum on $[1, 2]$, and the convergence in theorem 4.2 is uniform. At $y = 1$ the limit is one. Nonconstancy gives failure at some other mantissa. $\square$

A crude theorem-level bound follows without numerics: since $0 \leq F \leq 1$, $1/2 \leq \mathcal{P}(y) \leq 2$. Thus, after enlarging t_0 , one may for example take $A = 1/3$, $B = 3$ in (2).

4.3 Eventual behavior of the derivatives

For the regularity requested in the original question, the natural gauge has a useful fixed-order property.

Proposition 4.4 (fixed-order alternating derivatives). *For every fixed integer $m \geq 0$,*

$$g_1^{(m)}(x) = x^{-m} g_1(x) Q_m(\log x), \quad (30)$$

where Q_m is a degree- m polynomial with leading term $(-1)^m \lambda^{-m} X^m$. Consequently $(-1)^m g_1^{(m)}(x) > 0$ for all sufficiently large x , and each fixed derivative is eventually monotone.

Proof. Apply $d/dx = x^{-1} d/d(\log x)$ inductively to $g_1(x) = \exp[-L^2/(2\lambda) - \kappa_1 L]$, $L = \log x$. The leading term is multiplied at each step by $-L/\lambda$. Monotonicity of the m -th derivative follows from the eventual sign of the $(m + 1)$ -st. $\square$

The threshold depends on m ; this is not a claim of complete monotonicity on one common half-line.

5 What the singular profile does and does not forbid

The wave-2 audit correctly identified a strong obstruction, but stated its scope too broadly. We first formulate the exact obstruction, strengthen it, and then explain why it still leaves room for the adaptive construction in section 6.

5.1 Linearity forced by an unrestricted Cesàro limit

Lemma 5.1 (shell linearity). *If a positive gauge g satisfies (1), then for every $z \in [0, 1]$, uniformly in z ,*

$$2^{-k} \int_{2^k}^{2^{k(1+z)}} \frac{|\Phi(x)|}{g(x)} dx \longrightarrow z. \quad (31)$$

In particular, the normalized measures of $|\Phi|/g$ inside every dyadic shell converge weakly to Lebesgue measure on $[0, 1]$.

Proof. Apply (1) at 2^k and at $2^k(1+z)$, subtract the two asymptotics, and divide by 2^k . Uniformity follows from monotonicity of the partial integrals together with convergence on a fine fixed mesh. $\square$

5.2 Stabilizing and bounded-distortion gauges

Theorem 5.2 (stabilizing-shape obstruction). *Suppose there exist constants $c_k > 0$ and a continuous function $u : [0, 1] \rightarrow (0, \infty)$ such that*

$$\frac{g(2^k(1+z))}{c_k g_1(2^k(1+z))} \longrightarrow u(z) \quad \text{uniformly in } z. \quad (32)$$

Then g cannot satisfy (1).

Proof. Up to a scalar normalization, the shell measure for $|\Phi|/g$ is

$$u(z)^{-1} \alpha_k(dz)(1 + o(1)).$$

Every weak limit is therefore proportional to $u^{-1}\alpha$, which is singular. This contradicts the Lebesgue limit forced by [theorem 5.1](#). $\square$

The next statement removes the existence of any limiting shape.

Theorem 5.3 (bounded-distortion obstruction). *Suppose there are constants $0 < m < M < \infty$ and numbers $c_k > 0$ such that, for all sufficiently large k and all $z \in [0, 1]$,*

$$m \leq \frac{g(2^k(1+z))}{c_k g_1(2^k(1+z))} \leq M. \quad (33)$$

Then g cannot satisfy (1).

Proof. After cancelling the irrelevant scalar c_k , write the normalized shell measure as

$$\eta_k(dz) = \frac{q_k(z) \alpha_k(dz)}{\int q_k d\alpha_k}, \quad M^{-1} \leq q_k \leq m^{-1}.$$

Thus $\eta_k \leq C\alpha_k$ in the sense that $\int \varphi d\eta_k \leq C \int \varphi d\alpha_k$ for every continuous $\varphi \geq 0$, with C independent of k . Any weak limit η is therefore dominated by $C\alpha$, hence singular. Again this contradicts [theorem 5.1](#). $\square$

Corollary 5.4 (necessary shell microstructure). *Every gauge satisfying the unrestricted limit (1) must have unbounded shell distortion relative to g_1 , even after multiplication by an arbitrary scalar c_k on each shell.*

5.3 Hardy fields and the overreach about “elementary” functions

If g/g_1 belongs to a Hardy field, put $h = \log(g/g_1)$. Then $xh'(x)$ has a limit in $[-\infty, +\infty]$. A finite limit γ gives

$$h(2^k(1+z)) - h(2^k) \longrightarrow \gamma \log(1+z)$$

uniformly, so [theorem 5.2](#) applies. Infinite limits force shell mass to concentrate at one endpoint, contradicting (31). Hence no Hardy-field gauge of the correct rough scale can solve (1).

It does *not* follow that every elementary closed-form expression is covered. Trigonometric composition already leaves the Hardy-field world. For example, for $2c\varepsilon < 1/\lambda$,

$$g_{\varepsilon,c}(x) = g_1(x) \exp(\varepsilon \sin(c(\log x)^2)) \quad (34)$$

is elementary, positive, analytic, and eventually decreasing, but its shell shape does not stabilize. It still fails (1), now by [theorem 5.3](#), because its distortion is uniformly bounded. Thus the correct conclusion is:

Hardy-field, stabilizing, and bounded-distortion gauges are impossible. A blanket statement about all elementary formulas is not justified. The exact gauge constructed below escapes by developing unbounded, finer-and-finer shell structure.

6 An analytic decreasing gauge with the exact unrestricted Cesàro limit

This section resolves the literal existence question left open by all supplied materials. The proof has two stages. First we construct a smooth decreasing gauge by shellwise mass equalization. Then we approximate its logarithm and first derivative by an entire function with an arbitrarily small variable error.

6.1 A shellwise equalization lemma

Recall the measures α_k from (20). They converge to the continuous, full-support finite measure α .

Lemma 6.1 (smooth equalizers with a one-sided logarithmic slope). *There exist positive functions $q_k \in C^\infty([0, 1])$ such that:*

1. $q_k \equiv 1$ in neighborhoods of 0 and 1;
2. with $b_k = (k + \kappa_1 + 1)/4$,

$$(\log q_k)'(z) \geq -b_k \quad (0 \leq z \leq 1); \quad (35)$$

- 3.

$$\sup_{0 \leq z \leq 1} \left| \int_0^z q_k(u) \alpha_k(du) - z \right| \longrightarrow 0; \quad (36)$$

4. the construction can be arranged so that

$$\sup_{z \in [0,1]} |\log q_k(z)| = o(k). \quad (37)$$

Proof. Fix a partition depth r , write $\delta = 2^{-r}$, and let $I_{r,j} = [j\delta, (j+1)\delta]$, $0 \leq j < 2^r$. Since α has full support, $\alpha(I_{r,j}) > 0$ for every j . For fixed r , weak convergence and atomlessness give

$$\alpha_k(I_{r,j}) \longrightarrow \alpha(I_{r,j}) > 0$$

uniformly over the finite collection of cells. Define the step equalizer

$$q_{k,r}^{\text{step}}(z) = \frac{\delta}{\alpha_k(I_{r,j})} \quad (z \in I_{r,j}^\circ). \quad (38)$$

It gives exactly mass δ to each cell. Therefore its cumulative function differs from z by at most δ .

For fixed r , all the finitely many values in (38), and their reciprocals, remain bounded as $k \rightarrow \infty$. Replace the jumps by flat C^∞ bridges in small neighborhoods of the dyadic boundaries, interpolate in the logarithmic variable, and set the function equal to one near the two endpoints. Upward bridges have no constraint. A downward bridge with logarithmic drop D can be spread over width slightly larger than D/b_k , ensuring (35). Since $b_k \rightarrow \infty$, all bridge widths tend to zero for fixed r . The measures α_k converge to an atomless measure, so the total α_k -mass of these finitely many shrinking neighborhoods tends to zero. The smoothing therefore changes the cumulative integrals by $o(1)$, uniformly in the upper endpoint.

Now choose a diagonal sequence. For each r , take $K_r \geq r$ so large that all preceding estimates hold with error at most $1/r$, the bridges are disjoint, and the bounded logarithmic size C_r of the smoothed equalizer satisfies $C_r/K_r \leq 1/r$. For $K_r \leq k < K_{r+1}$, use depth r . Then $r \rightarrow \infty$, so $\delta + 1/r \rightarrow 0$, proving (36); the last choice of K_r proves (37). $\square$

The use of dyadic partition boundaries is especially natural here: for $r \leq k$, every interior partition boundary is a zero of P_{k+1} . The proof above only needs atomless weak convergence, but these exact zeros make the smoothing perturbation even smaller.

6.2 A smooth decreasing exact gauge

Theorem 6.2 (smooth exact Cesàro gauge). *There exists a positive C^∞ function g_0 , strictly decreasing on some half-line, such that for every fixed t_0 ,*

$$\frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{g_0(x)} dx \longrightarrow 1. \quad (39)$$

It can be chosen with

$$\log g_0(x) = -\frac{(\log x)^2}{2\lambda} - \kappa_1 \log x + o(\log x). \quad (40)$$

Proof. On the shell $x = 2^k(1+z)$, define

$$g_0(x) = \frac{g_1(x)}{q_k(z)}. \quad (41)$$

Because $q_k \equiv 1$ near both shell endpoints, these definitions join with all derivatives at powers of two; after defining the function arbitrarily on an initial compact interval, g_0 is globally smooth on a half-line.

Let $L = k\lambda + \log(1+z)$. Differentiating g_1 with respect to z gives

$$\frac{d}{dz} \log g_1(2^k(1+z)) = -\frac{k + \kappa_1 + \log_2(1+z)}{1+z}. \quad (42)$$

For $0 \leq z \leq 1$, the right side is at most $-(k + \kappa_1 + 1)/2 = -2b_k$. Hence (35) implies

$$\frac{d}{dz} \log g_0 = \frac{d}{dz} \log g_1 - (\log q_k)' \leq -b_k < 0$$

for all sufficiently large k ; thus g_0 is strictly decreasing.

By (19) and (41),

$$\frac{|\Phi(2^k(1+z))|}{g_0(2^k(1+z))} dz = q_k(z) \alpha_k(dz).$$

Consequently (36) gives, uniformly in z ,

$$\int_{2^k}^{2^{k+1}(1+z)} \frac{|\Phi(x)|}{g_0(x)} dx = 2^k(z + o(1)). \quad (43)$$

In particular a full shell contributes $2^k(1 + o(1))$. Summing the full shells before the terminal one and using the geometric Toeplitz argument yields

$$\int_{t_0}^t \frac{|\Phi(x)|}{g_0(x)} dx = t + o(t),$$

which is (39). Finally, (37) and $k \asymp \log x$ give (40). $\square$

6.3 Entire approximation preserving monotonicity

We use the following classical approximation theorem of Hoischen, in the modern form proved by Gauthier and Kienzle.

Theorem 6.3 (Hoischen–Carleman approximation). *Let $F \in C^1(\mathbb{R})$ and let $\varepsilon : \mathbb{R} \rightarrow (0, \infty)$ be continuous. There is an entire function G , real-valued on $\mathbb{R}$, such that*

$$|G(x) - F(x)| < \varepsilon(x), \quad |G'(x) - F'(x)| < \varepsilon(x) \quad (x \in \mathbb{R}). \quad (44)$$

Theorem 6.4 (analytic exact Cesàro gauge). *There exists a zero-free entire function $g(z) = \exp G(z)$, positive and strictly decreasing on a real half-line, such that its restriction to that half-line satisfies (1). Moreover it may be chosen so that*

$$\log g(x) = -\frac{(\log x)^2}{2 \log 2} - \kappa_1 \log x + o(\log x). \quad (45)$$

Proof. Extend $F = \log g_0$ from theorem 6.2 to a C^1 function on all of $\mathbb{R}$. On a sufficiently large half-line, $F'(x) < 0$. Choose a positive continuous error function with

$$\varepsilon(x) \longrightarrow 0, \quad \varepsilon(x) < -\frac{1}{2}F'(x)$$

on that half-line. Apply theorem 6.3. Then $G'(x) < F'(x)/2 < 0$, so $g(x) = e^{G(x)}$ is positive and strictly decreasing. Also

$$e^{-\varepsilon(x)} \leq \frac{g(x)}{g_0(x)} \leq e^{\varepsilon(x)}.$$

Since $\varepsilon(x) \rightarrow 0$, splitting the Cesàro integral at a fixed large point and squeezing the tail shows that replacing g_0 by g does not change the limit in (39). The logarithmic asymptotic follows from (40) and $|G - F| = o(1)$. $\square$

Remark 6.5 (constructive status). The shell equalizers are constructive once a numerical approximation to the finite measures α_k is available. The final entire approximation is an existence theorem rather than a compact elementary formula. This is not a cosmetic defect: theorem 5.4 proves that every exact gauge must acquire unbounded shell distortion. The simple natural formula cannot be repaired by a bounded periodic or slowly varying factor.

7 A natural exact repair: logarithmic averaging

The additive Cesàro mean gives the terminal dyadic shell a fixed positive fraction of the total interval. A logarithmic mean gives that shell weight $O(1/\log t)$, eliminating the phase profile.

Theorem 7.1 (unrestricted logarithmic mean). *Let*

$$A_1^{\log} = \frac{H_0}{\lambda} \nu\left(\frac{W}{1+\cdot}\right) = 0.09386063575679 \dots \quad (\text{numerical}). \quad (46)$$

Then for every fixed $t_0 > 0$,

$$\boxed{\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|\Phi(x)|}{A_1^{\log} g_1(x)} \frac{dx}{x} \rightarrow 1.} \quad (47)$$

Proof. The unnormalized mass of the k -th full logarithmic shell is

$$\beta_k = \int_{2^k}^{2^{k+1}} \frac{|\Phi(x)|}{g_1(x)} \frac{dx}{x} = \rho^{-(k+1)} \int_0^1 \frac{W(z)}{1+z} P_{k+1}(z) dz.$$

By (13), $\beta_k \rightarrow \beta_\infty = H_0 \nu(W/(1+\cdot))$, and the sequence is bounded. Up to $t \in [2^K, 2^{K+1})$, the numerator is the sum of $K + O(1)$ full shell masses plus a bounded terminal partial mass, while $\log(t/t_0) = K\lambda + O(1)$. Cesàro convergence of (β_k) gives (47) with $A_1^{\log} = \beta_\infty/\lambda$. $\square$

Thus the natural arithmetic formula has two exact interpretations: ordinary Cesàro convergence at dyadic endpoints, and unrestricted convergence for the multiplicatively invariant logarithmic mean.

8 Typical rays, RMS scale, and the corrected fluctuation law

8.1 The centered lacunary sum

Put

$$\psi(z) = \log |2 \sin \pi z|, \quad S_n(z) = \sum_{j=0}^{n-1} \psi(T^j z). \quad (48)$$

Since $\int_0^1 \psi = 0$, the Birkhoff sum is centered. From $P_n = 2^{-n} e^{S_n}$ and (5) one obtains, for fixed $1 < y < 2$,

$$\log |\Phi(2^k y)| = -\frac{\log^2(2^k y)}{2\lambda} - \kappa_0 \log(2^k y) + S_{k+1}(y-1) + \Psi(y), \quad (49)$$

where Ψ is bounded on compact subintervals of $(1, 2)$ and

$$\kappa_0 = \frac{3}{2} + \log_2 \pi. \quad (50)$$

Birkhoff's theorem immediately gives the typical-ray asymptotic.

Theorem 8.1 (typical dyadic ray). *For Lebesgue-almost every $y \in (1, 2)$,*

$$\log |\Phi(2^k y)| = -\frac{\log^2(2^k y)}{2 \log 2} - \kappa_0 \log(2^k y) + o(k). \quad (51)$$

The normalized remainder does not converge. Its exact variance and LIL can be computed without importing a delicate lacunary-series theorem.

8.2 Exact covariance and variance

In $L^2(0, 1)$,

$$\psi(z) = - \sum_{m=1}^{\infty} \frac{\cos(2\pi m z)}{m}. \quad (52)$$

Orthogonality gives, for $r \geq 0$,

$$\int_0^1 \psi(z) \psi(2^r z) dz = \frac{\pi^2}{12 \cdot 2^r}. \quad (53)$$

Therefore

$$\boxed{\int_0^1 S_n(z)^2 dz = \frac{\pi^2}{4} n - \frac{\pi^2}{3} (1 - 2^{-n})}. \quad (54)$$

In particular the asymptotic variance per dyadic step is $\pi^2/4$.

8.3 Martingale decomposition and LIL

Let P be the ordinary Perron operator of the doubling map,

$$(P\varphi)(x) = \frac{1}{2}(\varphi(x/2) + \varphi((x+1)/2)),$$

and define

$$d = 2\psi - \psi \circ T = \log |\tan \pi z|. \quad (55)$$

The Fourier series shows $Pd = 0$, while

$$\int_0^1 d(z)^2 dz = \frac{\pi^2}{4}. \quad (56)$$

Moreover the exact telescoping identity is

$$S_n = \sum_{j=0}^{n-1} d \circ T^j - \psi + \psi \circ T^n. \quad (57)$$

The first term is a reverse martingale sum. The logarithmic singularities of ψ have exponential tails, so the two coboundary terms are negligible on the LIL scale by Borel–Cantelli.

Theorem 8.2 (CLT and corrected LIL). *For Lebesgue-almost every z ,*

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N\left(0, \frac{\pi^2}{4}\right), \quad (58)$$

$$\limsup_{n \rightarrow \infty} \frac{S_n(z)}{\sqrt{2n \log \log n}} = \frac{\pi}{2}, \quad \liminf_{n \rightarrow \infty} \frac{S_n(z)}{\sqrt{2n \log \log n}} = -\frac{\pi}{2}. \quad (59)$$

Equivalently, with denominator $\sqrt{n \log \log n}$, the limsup constant is $\pi/\sqrt{2}$.

Proof. Apply the martingale central limit theorem and the martingale LIL to $\sum d \circ T^j$, using (56), then use (57). The tail estimate $\text{Leb}\{|\psi| > u\} = O(e^{-u})$ makes the coboundary terms $o(\sqrt{n \log \log n})$ almost surely. $\square$

Combining (49) and (59) yields the physical frequency form.

Corollary 8.3 (physical-ray LIL). *For almost every $y \in (1, 2)$,*

$$\limsup_{k \rightarrow \infty} \frac{\log |\Phi(2^k y)| + \frac{\log^2(2^k y)}{2\lambda} + \kappa_0 \log(2^k y)}{\sqrt{\log(2^k y) \log \log \log(2^k y)}} = \frac{\pi}{\sqrt{2\lambda}}, \quad (60)$$

with the negative of this constant as the liminf.

8.4 The $\sqrt{2}$ error in the published lacunary-product constant

The supplied Aistleitner–Hofer–Larcher source computes its variance by summing Fourier-coefficient products but omits the factor $1/2$ in $\int_0^1 \cos^2(2\pi jx) dx = 1/2$. It consequently prints $\sigma^2 = \pi^2/2$ instead of $\pi^2/4$. The exact finite- n formula (54) and the martingale identity independently force the smaller value.

Audit finding 8.4 (normalization of the correction). The correct constants are:

- $\pi/2$ with denominator $\sqrt{2n \log \log n}$;
- $\pi/\sqrt{2}$ with denominator $\sqrt{n \log \log n}$;
- $\pi/\sqrt{2 \log 2}$ in the physical-frequency normalization (60).

Each is smaller by $\sqrt{2}$ than the corresponding printed constant in the cited paper and in the wave-1 document that copied it.

8.5 The exact RMS exponent

For $q = 2$,

$$\int_0^1 P_n(z)^2 dz = 2^{-n}. \quad (61)$$

One proof is the identity $\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \mathbf{1}$; another is Fourier orthogonality. Thus $\rho_2 = 1/2$, $\Lambda_2 = \rho_2^{1/2} = 1/\sqrt{2}$, and the scale dictionary gives

$$\kappa_2 = \kappa(1/\sqrt{2}) = \log_2(2\pi). \quad (62)$$

This identifies the exponent in the deleted early conjecture: it is the root-mean-square exponent, not the typical, arithmetic, or extremal exponent.

9 The sharp extremal product and the coarse uniform envelope

Let $u_j = \sin^2(\pi 2^j z)$, so $u_{j+1} = 4u_j(1 - u_j)$. The following elementary subaction gives the exact Gelfond exponent.

Proposition 9.1 (sharp Gelfond bound). *For every $n \geq 1$ and $z \in \mathbb{R}$,*

$$P_n(z) \leq \sqrt{\frac{5}{3}} \left(\frac{\sqrt{3}}{2} \right)^n. \quad (63)$$

The exponential rate $\sqrt{3}/2$ is sharp and is attained at $z = 1/3, 2/3$.

Proof. Set $V(u) = 1 + 2u/3$. Direct expansion gives

$$3V(u) - 4uV(4u(1 - u)) = \frac{(2u + 1)(4u - 3)^2}{3} \geq 0. \quad (64)$$

Hence $u_j V(u_{j+1}) \leq (3/4)V(u_j)$. Multiplication yields

$$P_n(z)^2 V(u_n) \leq (3/4)^n V(u_0) \leq (3/4)^n \frac{5}{3},$$

which is (63). At $z = 1/3$ or $2/3$, every $u_j = 3/4$, so $P_n = (\sqrt{3}/2)^n$. $\square$

The scale dictionary now gives

$$\kappa_\infty = \kappa(\sqrt{3}/2) = \frac{3}{2} + \log_2 \frac{\pi}{\sqrt{3}}. \quad (65)$$

Theorem 9.2 (coarse global envelope). *There is a constant C such that, for all sufficiently large x ,*

$$|\Phi(x)| \leq CE(x)x^{-\kappa_\infty}. \quad (66)$$

The power exponent is sharp: on the ray $x = 2^k(4/3)$, the ratio of the two sides stays between positive constants.

Proof. Insert (63) into (15) with $\Lambda = \sqrt{3}/2$. Sharpness follows from the period-two orbit $z = 1/3$ and positivity of the bounded mantissa factor at $y = 4/3$. $\square$

Consequently

$$\limsup_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\frac{1}{2 \log 2}, \quad \liminf_{x \rightarrow \infty} \frac{\log |\Phi(x)|}{(\log x)^2} = -\infty, \quad (67)$$

where the second statement follows from the integer zeros.

10 Largest peak in a dyadic annulus: the Lambert- W law

The coarse envelope does not yet determine

$$M_k = \max_{2^k \leq x \leq 2^{k+1}} |\Phi(x)|.$$

A fixed maximizing ray such as $y = 4/3$ pays the linear denominator penalty $(4/3)^{-(k+1)}$. The true annular maximizer moves toward $y = 1$, paying only a sublinear boundary-layer loss while spending a slowly increasing number of doublings reaching the maximizing cycle.

Put

$$\beta = \log \frac{\sqrt{3}}{2}, \quad c = \frac{\lambda}{\pi} \sqrt{\frac{3}{2}} = 0.270222319733365 \dots \quad (68)$$

From (5), with $n = k + 1$,

$$M_k = 2^{-k(k+1)/2} \pi^{-n} A_n, \quad A_n = \sup_{0 \leq \theta \leq 1} H(1 + \theta)(1 + \theta)^{-n} P_n(\theta). \quad (69)$$

10.1 Boundary-layer reduction

Write a small θ as

$$\theta = 2^{-r} s, \quad s \in [1/4, 1/2].$$

The first r factors obey, uniformly in s ,

$$\sum_{j=0}^{r-1} \log |\sin(\pi 2^j \theta)| = -\frac{\lambda}{2} r^2 + r \left(\log(\pi s) - \frac{\lambda}{2} \right) + O(1). \quad (70)$$

The remaining orbit can be chosen to shadow the maximizing period-two cycle. The exact Gelfond upper bound and a rational-preimage lower construction give

$$\log P_{n-r}(s) = (n-r)\beta + O((\log \log n)^2) \quad (71)$$

for the optimization, uniformly at the required scale. The loss comes from a preimage depth $O(\log \log n)$.

After removing $n\beta$, the finite-dimensional objective is

$$\mathcal{F}_n(r, s) = -\frac{\lambda}{2} r^2 + r \left(\log(\pi s) - \frac{\lambda}{2} - \beta \right) - n \log(1 + s 2^{-r}). \quad (72)$$

10.2 Exact Lambert optimization

Set

$$C_0 = \log \pi - \frac{\lambda}{2} - \beta = \log \left(\pi \sqrt{\frac{2}{3}} \right), \quad A = C_0 + \log s, \quad z = \lambda r - A. \quad (73)$$

Replacing $-n \log(1 + s2^{-r})$ by $-ns2^{-r}$ changes the maximum by $o(1)$. A direct calculation gives

$$\mathcal{F}_n(r, s) = -\frac{z^2}{2\lambda} - ne^{-C_0}e^{-z} + \frac{A^2}{2\lambda} + o(1). \quad (74)$$

As s ranges through an interval of multiplicative width two, A ranges through an interval of length λ ; successive integer values of r therefore tile the real z -axis. The bounded A^2 -term is harmless. The remaining continuous maximum satisfies

$$ze^z = \lambda ne^{-C_0} = cn,$$

so $z = W_0(cn)$, the principal Lambert function.

Theorem 10.1 (sharp dyadic-annulus maximum). *Let $w_k = W_0(c(k+1))$. Then*

$$\boxed{\log M_k = -\frac{\lambda}{2}k(k+1) - (k+1)\log \pi + (k+1)\beta - \frac{w_k^2 + 2w_k}{2\lambda} + O((\log \log k)^2).} \quad (75)$$

Equivalently,

$$\begin{aligned} \log M_k = & -\frac{\lambda}{2}k^2 + (\beta - \log \pi - \lambda/2)k - \frac{(\log k)^2}{2\lambda} + \frac{\log k \log \log k}{\lambda} \\ & - \frac{1 + \log c}{\lambda} \log k + O((\log \log k)^2). \end{aligned} \quad (76)$$

A near-maximizer has

$$\frac{x}{2^k} = 1 + \Theta\left(\frac{W_0(c(k+1))}{k+1}\right), \quad r = \frac{W_0(c(k+1))}{\lambda} + O(1). \quad (77)$$

Proof. Equations (70)–(72), together with the sharp tail upper bound and the dense-preimage lower construction, reduce $\log A_n - n\beta$ to $\sup \mathcal{F}_n$ with error $O((\log \log n)^2)$. The tiling argument in (73)–(74) gives

$$\sup \mathcal{F}_n = -\frac{W_0(cn)^2 + 2W_0(cn)}{2\lambda} + O(1).$$

Insert this into (69). Expanding $W_0(cn) = \log n - \log \log n + \log c + O(\log \log n / \log n)$ gives (76). The critical point equation also gives $n\theta = W_0(cn)/\lambda$, proving (77). $\square$

The extra $-(\log k)^2/(2\lambda)$ term is a second log-normal suppression in $\log \log x$. It cannot be replaced by a fixed power of $\log x$.

11 Two-adic exceptional valleys and lobe maxima

Let E_m denote the unique maximum of $|\Phi|$ on $[m, m+1]$. Large powers of two create exceptionally deep valleys because many sinc factors vanish at the left endpoint.

Theorem 11.1 (peaks immediately after $m2^k$). *Fix an odd positive integer m , and write the unique maximizer in $(m2^k, m2^k + 1)$ as $m2^k + t_k$. Then*

$$(k+1)(1-t_k) \longrightarrow 1, \quad (78)$$

and

$$E_{m2^k} \sim \frac{H(1)|H(m)|}{e(k+1)} (m2^k)^{-(k+1)}. \quad (79)$$

In particular,

$$E_{2^k} \sim \frac{H(1)^2}{e(k+1)} 2^{-k(k+1)}, \quad H(1) = 0.553771275888810 \dots \quad (80)$$

Proof. Uniformly for $t \in [0, 1]$, the factors that vanish at $m2^k$ can be separated to give

$$|\Phi(m2^k + t)| = (m2^k)^{-(k+1)} |H(m)| t^{k+1} \Phi(t)(1 + o(1)). \quad (81)$$

As $t \uparrow 1$, $\Phi(t) = H(1)(1-t)(1+o(1))$. Thus the remaining maximization is the standard boundary problem $t^p(1-t)$, $p = k+1$, whose maximizer satisfies $p(1-t) \rightarrow 1$ and whose maximum is asymptotic to $1/(ep)$. Strict uniqueness follows from [theorem 2.3](#). $\square$

Comparing (80) with (75) gives

$$\log \frac{E_{2^k}}{M_k} = -\frac{\lambda}{2} k^2 + O(k),$$

so the first lobe after a power of two is vastly smaller than the largest lobe in the same dyadic annulus.

12 Numerical adjudication and reproducibility

All theorem-level conclusions above are independent of the decimal values in this section. The numerics are useful for checking normalizations and for correcting two coarse values in the comparative audit.

12.1 Spectral and shell constants

Chebyshev–Lobatto collocation of $\mathcal{L}$, with barycentric evaluation at the two inverse branches, is stable from 32 to 80 nodes and gives

$$\begin{aligned} \rho &= 0.661322602060565 \dots, \\ \kappa_1 &= 2.748070014871334 \dots, \\ A_1 &= 0.09126612413159 \dots, \\ A_1^{\log} &= 0.09386063575679 \dots \end{aligned} \quad (82)$$

An independent direct quadrature on all 2^{18} dyadic cells, using 16-point Gauss–Legendre quadrature in each cell, gives

$$A_1^{(18)} = 0.09126612413156757, \quad (A_1^{\log})^{(18)} = 0.09386063575678312.$$

12.2 Corrected profile range

The comparative audit reported a range $[0.9249, 1.1133]$, explicitly on a mantissa grid of spacing 0.025. That grid misses the true narrow excursions. The same $n = 18$ cell quadrature, with within-cell root refinement for $\mathcal{P}'_n$, gives

quantity	value	location
finite-level minimum	0.9162685832208	$y \approx 1.142840221249$
finite-level maximum	1.1181397445438	$y \approx 1.428632869969$

At level $n = 19$, the dyadic-cell boundary values are already 0.9162686675 and 1.1181395810, supporting convergence to nearby limits. These are high-resolution numerical estimates, not certified exact constants for the limiting singular profile.

12.3 Minimal reproducibility pseudocode

Listing 1: Direct dyadic-cell quadrature for the natural profile

```
rho = 0.661322602060565
lam = log(2)
kappa = 0.5 + log(pi/rho)/lam
n = 18
N = 2**n

for each dyadic cell I_j = [j/N, (j+1)/N]:
    use 16-point Gauss-Legendre nodes z in I_j
    logP = sum_{ell=0}^{n-1} log(abs(sin(pi * frac(2**ell*z))))
    logH = sum_{m=1}^{55} log(sin(pi*(1+z)/2**m)/(pi*(1+z)/2**m))
    logW = log(rho/pi) + logH + (kappa-1)*log(1+z)
           + log(1+z)**2/(2*lam)
    density = exp(-n*log(rho) + logW + logP)
    cell_mass[j] = Gauss_integral(density, I_j)

A1 = sum(cell_mass)
F_at_boundaries = cumulative_sum(cell_mass/A1)
profile = (1 + F_at_boundaries)/(1 + arange(N+1)/N)
```

The exact variance (54) supplies an especially strong independent check: at $n = 1000$ it gives 2464.1112321386..., asymptotically half the value implied by the erroneous $\pi^2/2$ variance.

13 Document-by-document adjudication

source		strongest correct contribution	correction or limitation
Wave-1 document 1		Exact dyadic factorization; typical-ray exponent; covariance, variance $\pi^2/4$, and the correct LIL normalization; coarse extremal analysis.	Does not solve the arithmetic Cesàro problem. Its annular expansion is correct only to a coarser remainder than document 2.
Wave-1 document 2		Sharp Gelfond subtraction; unique lobe peak; full Lambert- W annular asymptotic; two-adic peak theorem.	Deliberately follows the pointwise/extremal track and does not determine the arithmetic transfer eigenvalue or the original Cesàro gauge.

source		strongest correct contribution	correction or limitation
Wave-1 document 3	docu-	Broad synthesis; correct separation of typical, arithmetic, RMS, and extremal exponents; construction of the natural analytic gauge satisfying the bounded requirement.	Inherits the published Aistleitner–Hofer–Larcher LIL constant, which is too large by $\sqrt{2}$. Its discussion of failure of the all-time limit applies to the natural gauge, not to all possible analytic gauges.
Wave-1 document 4	docu-	Most systematic transfer-operator and L^q treatment; correct arithmetic exponent, RMS identity, singular-profile mechanism, and bounded answer.	The spectral-gap package is imported rather than fully proved; fluctuation theory and the sharp Lambert correction are not developed. Conditional intermediate-rate heuristics should not be read as established theorems.
Aistleitner– Hofer–Larcher		Identifies the Fouvry–Mauduit arithmetic constant and proves the rigorous enclosure $0.66130 < \rho < 0.66135$; supplies deep lacunary-product estimates.	Its variance computation omits the Parseval factor $1/2$, printing $\pi^2/2$ instead of $\pi^2/4$; affected LIL constants must be divided by $\sqrt{2}$.
Wave-2 compar- ative audit		Correctly reconciles the exponent spectrum, proves singularity of the arithmetic eigenmeasure, identifies the natural phase profile, and introduces the exact logarithmic-mean repair.	The reported profile range is a coarse-grid range. The implication “Hardy-field, hence every elementary formula” is too broad. Most importantly, “unattainable in principle” is false without a bounded- or stabilizing-distortion hypothesis: theorem 6.4 gives an unrestricted analytic exact gauge.

14 Final theorem and practical interpretation

The results can be compressed into one theorem that answers every interpretation appearing in the supplied materials.

Theorem 14.1 (complete Fourier-decay resolution). *Let $\Phi(x) = \prod_{n \geq 0} \text{sinc}(\pi x/2^n)$. Then:*

1. *There is no positive global pointwise asymptotic equivalent. The universal log-square limsup is $-1/(2 \log 2)$, while integer zeros force liminf $-\infty$.*
2. *For almost every dyadic mantissa, the power correction is $\kappa_0 = 3/2 + \log_2 \pi$, with CLT variance $\pi^2/4$ and physical LIL constant $\pi/\sqrt{2 \log 2}$.*
3. *The natural arithmetic gauge is $g_{\text{nat}} = A_1 E(x) x^{-\kappa_1}$, with $\kappa_1 = 1/2 + \log_2(\pi/\rho)$. It satisfies the bounded condition (2); its unrestricted limit points form the range of $\mathcal{P}(y) = (1 + F(y-1))/y$, and it gives limit one at dyadic endpoints.*
4. *No gauge with bounded shell distortion relative to g_1 , no stabilizing-shell gauge, and no Hardy-field gauge of the correct scale can satisfy the exact unrestricted limit (1).*
5. *Nevertheless, there exists a positive real-analytic, eventually strictly decreasing gauge satisfying (1). It can have the same rough log-normal–power scale as g_1 , but must develop unbounded fine shell microstructure.*
6. *The natural gauge also gives an exact unrestricted logarithmic mean after multiplication by $A_1^{\log}$.*

7. The RMS power is $\kappa_2 = \log_2(2\pi)$; the coarse uniform power is $\kappa_\infty = 3/2 + \log_2(\pi/\sqrt{3})$.
8. The largest dyadic-annulus peak obeys the Lambert-W formula (75); lobe peaks after $m2^k$ obey (79).

For applications, the choice of answer is therefore straightforward.

- Use $A_1 g_1$ for a simple analytic arithmetic envelope, two-sided Cesàro control, or dyadic-endpoint normalization.
- Use $A_1^{\log} g_1$ when multiplicative/logarithmic averaging is natural and an unrestricted exact limit is desired.
- Use the adaptive analytic gauge of [theorem 6.4](#) only when the literal additive Cesàro limit is essential; it is necessarily noncanonical.
- Use κ_0 , κ_2 , or κ_∞ only for the specific typical, RMS, or extremal notion they encode.
- Use the Lambert formula, not a fixed logarithmic power, for the largest peak in a dyadic annulus.

The phrase “how fast does the Fourier transform decay?” has one universal first answer and several incompatible second answers. Universally, the quadratic logarithmic coefficient is $-1/(2 \log 2)$. Beyond that order, the correct exponent and correction depend on whether one asks for typical values, means, RMS size, a coarse envelope, an annular maximum, or a specified exceptional lobe. The original additive Cesàro existence question is yes, but only through a deliberately adaptive analytic gauge; the simple canonical gauge has a singular nonconstant phase profile instead.

A Details behind the exponent spectrum

For $q > 0$, let ρ_q be the leading eigenvalue of $\mathcal{L}_q$ and set

$$\Lambda_q = \rho_q^{1/q}, \quad \kappa_q = \frac{\log \pi + \lambda/2 - \log \Lambda_q}{\lambda}. \quad (83)$$

Then the q -th moment of the normalized shell ratio $|\Phi|/(Ex^{-\kappa_q})$ has a finite nonzero limit under the corresponding RPF hypotheses. At the endpoints:

- $q \downarrow 0$: $\log \Lambda_0 = \int_0^1 \log |\sin \pi z| dz = -\log 2$, giving κ_0 .
- $q = 1$: $\Lambda_1 = \rho$, giving κ_1 .
- $q = 2$: $\rho_2 = 1/2$, hence $\Lambda_2 = 1/\sqrt{2}$, giving κ_2 .
- $q \rightarrow \infty$: $\Lambda_\infty = \sqrt{3}/2$, giving κ_∞ .

The map $q \mapsto \log \rho_q$ is convex by Hölder’s inequality, which organizes the monotone interpolation between these distinguished values.

B Why the adaptive construction does not contradict singularity

For the natural gauge, the normalized shell measures converge to the singular measure μ . A bounded multiplier cannot change the measure class, which is exactly [theorem 5.3](#). The equalizers q_k in [theorem 6.1](#) do something qualitatively different. At partition depth r , they assign reciprocal mass

$$q_{k,r}|_{I_{r,j}} \approx \frac{2^{-r}}{\alpha_k(I_{r,j})}.$$

As $r \rightarrow \infty$, singularity forces these reciprocals to have unbounded relative variation. The depth grows only after k is large enough that the required downward logarithmic bridges fit inside the cells while respecting monotonicity of g_1/q_k . Thus the construction converges weakly to Lebesgue at every fixed resolution, but has no bounded limiting density with respect to μ . That is precisely the escape route left open by the singular-profile theorem.

C Source map and bibliographic notes

The repository paths below are included so that every audited claim can be traced to its source. The present article does not assume the correctness of those drafts; the proofs above adjudicate their overlapping statements.

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